Student Name:Seang Sony

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Weekly Report

Summary of work completed:

1. Built a 7-stage image preprocessing pipeline (noise reduction, contrast fix, deskew).
2. Added auto-detection to apply different processing to clean PDFs vs. bad photos.
3. Wrote a 4-step post-processing script to fix Tesseract's common Khmer character errors.
4. Increased OCR confidence from 77.2% to 78.1% on test documents.

In Progress:

1. Testing the new pipeline on a wider variety of Khmer PDFs to ensure it works consistently.
2. Documenting edge cases where stacked consonants (ជើង) still fail in Tesseract.

Plans for Next Week:

1. Implement Google Gemini AI to correct leftover garbage text that Python rules can't fix.
2. Measure accuracy using formal Character Error Rate (CER) and Word Error Rate (WER) metrics.
3. Research layout analysis to crop out logos and page numbers before OCR.

Challenges & Issues:

1. Tesseract still struggles heavily with thin Khmer vowels and complex clusters.
2. Heavy image cleaning fixes bad photos but actively ruins clean digital PDFs, making auto-detection tricky.